County matching: Glaeser and Gottlieb (2009)

Control: Match of Historical to Modern Geographic Units — Glaeser and Gottlieb (2009). At State level, this is a historical-instrument choice.

1. Match county identities. This option matches historical county identities to current county units, using county codes and explicit identity corrections. Counties then enter their current MSA or, in State analysis, the historical FE group assigned in the dataset for the territory year. The regional membership is fixed when the instrument year changes. This is the project's Glaeser–Gottlieb-style county matching.

2. Add whole historical populations. Let P_{jt} be the population of historical county j in census year t, and let $m_{jg,t} = 1$ when that county is assigned to target region g. Otherwise, $m_{jg,t} = 0$. The matched population is

$$P_{g,t}^{\text{match}} = \sum_{j} m_{jg,t} P_{jt}.$$

Under Levels, the instrument is $z_{g,t} = P_{g,t}^{\text{match}}/1{,}000$. It measures thousands of people. Natural log instead uses $\log P_{g,t}^{\text{match}}$ for positive totals. The population total is used with both the employment and employment-density approaches. The minimum-overlap control does not apply to this method.

3. Matching does not divide population by shared area. If a historical county later splits, the match can assign its whole population to one identified successor. It does not distribute that population among all modern descendants. If all descendants remain in the same MSA, the MSA sum can nevertheless be unchanged. If a split crosses the modern MSA's boundary, matching and area allocation can produce different totals. The county identities, rather than the proportion of land inside the modern region, determine the assignment.

4. Unresolved sources remain a limitation. For State estimates, sum the recorded populations that can be assigned to the region with confidence. An unresolved identity or an unknown population leaves that source out of the sum. A verified joint report counts once when all its members belong to the region. If the members span regions and their separate populations are unknown, omit the joint report. The regional sum can therefore cover only part of the population. The State coverage selector applies the same 80%, 95%, or no minimum to the share of land with a recorded population in both historical years. This percentage measures geographic coverage. County and MSA totals follow their own source completeness checks.